StudySapiens Board MCQs

Chapter 12 — Human Skeletal and Muscular Systems

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. The internal framework of the body is called the:

(a) Skin
(b) Fascia only
(c) Muscle
(d) Skeleton

2. The membrane covering the surface of bone is the:

(a) Perichondrium
(b) Pericardium
(c) Pleura
(d) Periosteum

3. The hard outer layer of bone is called:

(a) Cartilage
(b) Marrow only
(c) Compact bone
(d) Spongy bone

4. The basic structural unit of compact bone is the:

(a) Nephron
(b) Alveolus
(c) Sarcomere
(d) Haversian system

5. Concentric layers of bone matrix are called:

(a) Trabeculae only
(b) Lamellae
(c) Lacunae
(d) Canaliculi

6. Mature bone cells are called:

(a) Chondrocytes
(b) Osteoclasts
(c) Osteoblasts
(d) Osteocytes

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7. Bone-forming cells are called:

(a) Osteoblasts
(b) Osteoclasts
(c) Osteocytes
(d) Chondrocytes

8. Bone-resorbing cells are called:

(a) Osteocytes
(b) Fibroblasts
(c) Osteoblasts
(d) Osteoclasts

9. Small spaces containing osteocytes are called:

(a) Haversian canals
(b) Lacunae
(c) Canaliculi
(d) Volkmann canals only

10. Canaliculi connect:

(a) Osteocytes to each other and Haversian canals
(b) Muscle fibres only
(c) Cartilage cells only
(d) Neurons only

11. The Haversian canal contains:

(a) Only marrow fat
(b) Blood vessels, nerves and lymphatics
(c) Only cartilage
(d) Only synovial fluid

12. Red bone marrow produces:

(a) Insulin only
(b) Blood cells
(c) Bile only
(d) Synovial fluid only

13. Yellow bone marrow mainly stores:

(a) Red blood cells only
(b) Calcium only
(c) Glucose only
(d) Fat

14. The process of bone formation is called:

(a) Osteogenesis
(b) Bone resorption
(c) Hematopoiesis
(d) Articulation

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15. Long bones develop by replacement of:

(a) Cartilage with bone
(b) Blood with bone
(c) Muscle with bone
(d) Fat with bone

16. Cartilage is surrounded by the:

(a) Pleura
(b) Periosteum
(c) Peritoneum
(d) Perichondrium

17. Cartilage cells are called:

(a) Chondrocytes
(b) Osteocytes
(c) Osteoblasts
(d) Myocytes

18. Cartilage heals slowly because it:

(a) Contains marrow
(b) Is made of bone
(c) Has many nerves
(d) Lacks blood vessels

19. The most common type of cartilage is:

(a) Elastic cartilage
(b) Hyaline cartilage
(c) Fibrocartilage
(d) Calcified cartilage only

20. Intervertebral discs contain:

(a) Elastic cartilage only
(b) Fibrocartilage
(c) Compact bone only
(d) Hyaline cartilage only

21. The external ear contains:

(a) Compact bone only
(b) Fibrocartilage only
(c) Hyaline cartilage only
(d) Elastic cartilage

22. The human skeleton contains how many bones?

(a) 300
(b) 206
(c) 210
(d) 200

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23. The axial skeleton includes the:

(a) Limbs only
(b) Pectoral girdle only
(c) Skull, vertebral column, ribs and sternum
(d) Pelvic girdle only

24. The appendicular skeleton includes the:

(a) Ribs only
(b) Skull only
(c) Vertebral column only
(d) Limbs and girdles

25. The skull contains how many bones?

(a) 33
(b) 8
(c) 22
(d) 14

26. The number of cranial bones is:

(a) 8
(b) 22
(c) 14
(d) 12

27. The number of facial bones is:

(a) 8
(b) 14
(c) 12
(d) 22

28. The middle ear ossicles are:

(a) Atlas, axis and hyoid
(b) Malleus, incus and stapes
(c) Scapula, clavicle and sternum
(d) Radius, ulna and humerus

29. The hyoid bone is unique because it:

(a) Is a rib
(b) Is part of the vertebral column
(c) Does not articulate with any other bone
(d) Is a facial bone

30. The vertebral column contains how many vertebrae?

(a) 26
(b) 33
(c) 30
(d) 24

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31. The number of cervical vertebrae is:

(a) 4
(b) 12
(c) 5
(d) 7

32. The first cervical vertebra is called the:

(a) Hyoid
(b) Atlas
(c) Sacrum
(d) Axis

33. The second cervical vertebra is called the:

(a) Coccyx
(b) Axis
(c) Hyoid
(d) Atlas

34. Rib-bearing vertebrae are the:

(a) Lumbar vertebrae
(b) Cervical vertebrae
(c) Sacral vertebrae
(d) Thoracic vertebrae

35. The sacrum is formed by fusion of:

(a) Four coccygeal vertebrae
(b) Five sacral vertebrae
(c) Seven cervical vertebrae
(d) Twelve thoracic vertebrae

36. True ribs are the first:

(a) 12 pairs
(b) 10 pairs
(c) 8 pairs
(d) 7 pairs

37. Floating ribs are the:

(a) 11th and 12th pairs
(b) 1st–7th pairs
(c) All 12 pairs
(d) 8th–10th pairs

38. Joints that allow little or no movement are:

(a) Hinge joints
(b) Ball-and-socket joints
(c) Synovial joints
(d) Fibrous joints

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39. The pubic symphysis is a:

(a) Cartilaginous joint
(b) Synovial joint
(c) Fibrous joint
(d) Ball-and-socket joint

40. The most common and freely movable joints are:

(a) Sutures only
(b) Synovial joints
(c) Fibrous joints
(d) Cartilaginous joints only

41. Synovial fluid is secreted by the:

(a) Periosteum only
(b) Synovial membrane
(c) Articular cartilage only
(d) Ligaments only

42. The shoulder and hip joints are:

(a) Pivot joints
(b) Hinge joints
(c) Gliding joints
(d) Ball-and-socket joints

43. The elbow and knee joints are mainly:

(a) Hinge joints
(b) Ball-and-socket joints
(c) Saddle joints
(d) Pivot joints

44. The joint between atlas and axis is a:

(a) Saddle joint
(b) Pivot joint
(c) Ball-and-socket joint
(d) Hinge joint

45. The thumb joint is a:

(a) Saddle joint
(b) Gliding joint only
(c) Pivot joint
(d) Hinge joint

46. A herniated disc occurs when:

(a) Bone fuses with muscle
(b) Cartilage becomes bone
(c) The inner disc material leaks and presses on nerves
(d) Synovial fluid is produced

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47. Spondylosis refers to:

(a) Joint dislocation only
(b) Muscle inflammation
(c) Bone infection only
(d) Degeneration of vertebrae/discs

48. Sciatica involves irritation of the:

(a) Vagus nerve
(b) Sciatic nerve
(c) Phrenic nerve only
(d) Optic nerve

49. Osteoarthritis is caused by:

(a) Autoimmune attack on synovium only
(b) Bacterial infection only
(c) Degeneration of articular cartilage
(d) Uric acid crystals only

50. Rheumatoid arthritis is mainly:

(a) An autoimmune inflammation of synovial membrane
(b) Bone infection only
(c) Uric acid deposition only
(d) Cartilage wear only

51. Gouty arthritis is caused by deposition of:

(a) Bile salts only
(b) Uric acid crystals
(c) Calcium only
(d) Cholesterol only

52. Osteoporosis is characterized by:

(a) Increased bone density
(b) Muscle wasting only
(c) Joint swelling only
(d) Decreased bone density and fragile bones

53. A common cause of osteoporosis in women is:

(a) Increase in calcium
(b) Excess exercise only
(c) Drop in oestrogen after menopause
(d) High vitamin D only

54. Vitamin D is important for bones because it helps absorb:

(a) Iron only
(b) Sodium only
(c) Potassium only
(d) Calcium

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55. Dislocation is:

(a) Tearing of ligaments only
(b) Bone fracture only
(c) Muscle spasm only
(d) Displacement of bones from a joint

56. A sprain is injury to:

(a) Bones only
(b) Muscles only
(c) Cartilage only
(d) Ligaments

57. Muscle tissue is specialized for:

(a) Secretion only
(b) Photosynthesis
(c) Filtration
(d) Contraction

58. Skeletal muscles are:

(a) Involuntary and smooth
(b) Voluntary and non-striated
(c) Involuntary and striated
(d) Voluntary and striated

59. Smooth muscles are:

(a) Voluntary and striated
(b) Involuntary and non-striated
(c) Involuntary and striated
(d) Voluntary and non-striated

60. Cardiac muscle is:

(a) Voluntary and branched
(b) Voluntary and smooth
(c) Involuntary, striated and branched
(d) Involuntary and non-striated

61. The plasma membrane of a muscle fibre is called the:

(a) Perimysium
(b) Plasmalemma only
(c) Sarcolemma
(d) Tonoplast

62. The cytoplasm of a muscle fibre is called:

(a) Nucleoplasm
(b) Stroma
(c) Sarcoplasm
(d) Matrix

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63. The endoplasmic reticulum of muscle cells is called:

(a) Pericardium
(b) Golgi apparatus
(c) Tonoplast
(d) Sarcoplasmic reticulum

64. T-tubules are invaginations of the:

(a) Sarcoplasmic reticulum only
(b) Golgi membrane
(c) Sarcolemma
(d) Nuclear envelope

65. Myofibrils are composed of:

(a) Tubulin only
(b) Collagen only
(c) Keratin only
(d) Actin and myosin filaments

66. The thick filaments are made of:

(a) Troponin
(b) Tropomyosin
(c) Myosin
(d) Actin

67. The thin filaments are mainly made of:

(a) Myosin
(b) Keratin
(c) Actin
(d) Collagen

68. The functional unit of muscle contraction is the:

(a) Sarcomere
(b) Nephron
(c) Neuron
(d) Osteon

69. The dark band in a sarcomere is the:

(a) H-zone
(b) A-band
(c) Z-line
(d) I-band

70. The light band in a sarcomere is the:

(a) I-band
(b) M-line
(c) A-band
(d) H-zone

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71. During contraction, the H-zone and I-band:

(a) Narrow
(b) Disappear only in A-band
(c) Widen
(d) Remain unchanged

72. During contraction, the A-band:

(a) Shortens
(b) Widens
(c) Remains the same length
(d) Disappears

73. Calcium ions bind to _____ during muscle contraction.

(a) Troponin
(b) Tropomyosin only
(c) Actin only
(d) Myosin

74. In relaxed muscle, myosin-binding sites on actin are blocked by:

(a) Myosin heads
(b) Tropomyosin
(c) ATP only
(d) Troponin only

75. The power stroke occurs when myosin heads:

(a) Pull actin filaments toward the centre
(b) Bind ATP only
(c) Detach from actin
(d) Release calcium

76. ATP binding to myosin causes:

(a) Detachment of myosin from actin
(b) Calcium release only
(c) Power stroke only
(d) Troponin binding only

77. Rigor mortis occurs because after death there is no:

(a) Calcium only
(b) Oxygen only
(c) Glucose only
(d) ATP to detach myosin from actin

78. Muscles that oppose each other's action are called:

(a) Synergistic muscles only
(b) Extensors only
(c) Antagonistic muscles
(d) Flexors only

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79. Flexion of the knee is produced by the:

(a) Quadriceps
(b) Biceps only
(c) Hamstrings
(d) Triceps only

80. Extension of the knee is produced by the:

(a) Hamstrings
(b) Deltoid only
(c) Quadriceps
(d) Gastrocnemius only

81. The stationary attachment of a muscle is called the:

(a) Belly
(b) Tendon only
(c) Insertion
(d) Origin

82. The movable attachment of a muscle is called the:

(a) Origin
(b) Fascia only
(c) Belly
(d) Insertion

83. Muscle fatigue may result from:

(a) High pH only
(b) Excess oxygen only
(c) ATP depletion and metabolite accumulation
(d) Increased calcium only

84. Muscle cramps are often associated with:

(a) Dehydration and salt imbalance
(b) Excess calcium only
(c) Low temperature only
(d) High oxygen only

85. Tetany is commonly caused by:

(a) Low oxygen only
(b) High blood glucose
(c) Low blood calcium
(d) High blood calcium

86. The S-shaped curve of the human vertebral column helps:

(a) Produce blood cells
(b) Store fat only
(c) Distribute weight and maintain balance
(d) Increase flexibility only

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87. The human pelvis is adapted for bipedalism by being:

(a) Absent
(b) Fused to the skull
(c) Longer and narrower
(d) Shorter and broader

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ANSWER KEY WITH BRIEF REASONS

1. (d) Skeleton — Bones/cartilage form the skeleton.

2. (d) Periosteum — Periosteum covers bone.

3. (c) Compact bone — Compact bone is dense.

4. (d) Haversian system — Osteons/Haversian systems make compact bone.

5. (b) Lamellae — Lamellae surround the Haversian canal.

6. (d) Osteocytes — Osteocytes maintain bone.

7. (a) Osteoblasts — Osteoblasts secrete matrix.

8. (d) Osteoclasts — Osteoclasts break down bone.

9. (b) Lacunae — Lacunae house bone cells.

10. (a) Osteocytes to each other and Haversian canals — They allow nutrient exchange.

11. (b) Blood vessels, nerves and lymphatics — It supplies bone tissue.

12. (b) Blood cells — It is hematopoietic.

13. (d) Fat — It is an energy reserve.

14. (a) Osteogenesis — Osteogenesis is the formation of bone.

15. (a) Cartilage with bone — Endochondral ossification forms long bones.

16. (d) Perichondrium — Perichondrium covers cartilage.

17. (a) Chondrocytes — Chondrocytes maintain cartilage matrix.

18. (d) Lacks blood vessels — Nutrients arrive by diffusion.

19. (b) Hyaline cartilage — Hyaline occurs at joints/trachea/nose.

20. (b) Fibrocartilage — Fibrocartilage resists stress.

21. (d) Elastic cartilage — Elastic cartilage provides flexibility.

22. (b) 206 — Adult skeleton has 206 bones.

23. (c) Skull, vertebral column, ribs and sternum — It forms the body axis.

24. (d) Limbs and girdles — It supports appendages.

45. (a) Saddle joint — It allows movement in two planes.

46. (c) The inner disc material leaks and presses on nerves — It causes pain/numbness.

47. (d) Degeneration of vertebrae/discs — It is age-related degeneration.

48. (b) Sciatic nerve — Pain radiates down the leg.

49. (c) Degeneration of articular cartilage — It is wear-and-tear arthritis.

50. (a) An autoimmune inflammation of synovial membrane — Immune system attacks joints.

51. (b) Uric acid crystals — Crystals inflame joints.

52. (d) Decreased bone density and fragile bones — Bones become porous.

53. (c) Drop in oestrogen after menopause — Hormonal change accelerates bone loss.

54. (d) Calcium — Calcium is needed for bone strength.

55. (d) Displacement of bones from a joint — Joint surfaces lose normal alignment.

56. (d) Ligaments — Ligaments are stretched/torn.

57. (d) Contraction — Muscles generate force/movement.

58. (d) Voluntary and striated — They move the skeleton.

59. (b) Involuntary and non-striated — They occur in internal organs.

60. (c) Involuntary, striated and branched — It forms the heart wall.

61. (c) Sarcolemma — Sarcolemma encloses the muscle cell.

62. (c) Sarcoplasm — It contains myofibrils.

63. (d) Sarcoplasmic reticulum — It stores calcium.

64. (c) Sarcolemma — They conduct impulses into the fibre.

65. (d) Actin and myosin filaments — Thin/thick filaments produce contraction.

66. (c) Myosin — Myosin forms thick filaments.

67. (c) Actin — Actin forms thin filaments.

68. (a) Sarcomere — Sarcomere lies between Z-lines.

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25. (c) 22 — 8 cranial + 14 facial bones.

26. (a) 8 — Cranium has 8 bones.

27. (b) 14 — Face has 14 bones.

28. (b) Malleus, incus and stapes — Three bones transmit sound.

29. (c) Does not articulate with any other bone — It is suspended in the neck.

30. (b) 33 — 7 cervical + 12 thoracic + 5 lumbar + 5 sacral + 4 coccygeal.

31. (d) 7 — Neck has 7 vertebrae.

32. (b) Atlas — Atlas supports the skull.

33. (b) Axis — Axis allows rotation.

34. (d) Thoracic vertebrae — 12 thoracic vertebrae articulate with ribs.

35. (b) Five sacral vertebrae — Sacral vertebrae fuse.

36. (d) 7 pairs — They attach directly to sternum.

37. (a) 11th and 12th pairs — They do not attach anteriorly to sternum.

38. (d) Fibrous joints — Fibrous joints are immovable/slightly movable.

39. (a) Cartilaginous joint — It is joined by fibrocartilage.

40. (b) Synovial joints — Synovial joints allow wide motion.

41. (b) Synovial membrane — It lubricates the joint.

42. (d) Ball-and-socket joints — They allow multiaxial movement.

43. (a) Hinge joints — They move mainly in one plane.

44. (b) Pivot joint — It allows rotation.

69. (b) A-band — A-band contains thick filaments.

70. (a) I-band — I-band contains thin filaments only.

71. (a) Narrow — Thin filaments slide inward.

72. (c) Remains the same length — Thick filament length does not change.

73. (a) Troponin — Troponin moves tropomyosin.

74. (b) Tropomyosin — Tropomyosin covers binding sites.

75. (a) Pull actin filaments toward the centre — Conformational change slides filaments.

76. (a) Detachment of myosin from actin — ATP is needed for cross-bridge cycling.

77. (d) ATP to detach myosin from actin — Cross-bridges remain locked.

78. (c) Antagonistic muscles — They produce opposite movements.

79. (c) Hamstrings — Hamstrings bend the leg.

80. (c) Quadriceps — Quadriceps straighten the leg.

81. (d) Origin — Origin remains fixed during contraction.

82. (d) Insertion — Insertion moves when muscle contracts.

83. (c) ATP depletion and metabolite accumulation — Metabolic changes reduce performance.

84. (a) Dehydration and salt imbalance — Electrolyte imbalance triggers spasms.

85. (c) Low blood calcium — Hypocalcaemia increases excitability.

86. (c) Distribute weight and maintain balance — It supports bipedal posture.

87. (d) Shorter and broader — It stabilizes the torso.

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